THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{7}{9}$	Each is $\frac{d}{9}$.
2	**(b)** $\frac{4}{9}$	Add the numerators over 99.
3	**(a)** 3	Cancel every prime factor other than 2 and 5.
4	**(b)** 4	$\frac{1}{x} = a - \sqrt{3}$.
5	**(c)** $\frac{14}{11}$	Write as $1 + 0.\overline{ab}$.
6	**(a)** 1	Reason about decimal expansions.
7	**(b)** 5	Find the order of 10 modulo q.
8	**(a)** 2	Multiply the fraction by 11 and reduce.
9	**(b)** $\frac{4}{3}$	Each is $\frac{d}{9}$.
10	**(c)** $\frac{8}{9}$	Add the numerators over 99.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$0.\overline{17} = \frac{17}{99}$, $0.\overline{71} = \frac{71}{99}$ (1); Sum $= \frac{8}{9}$ (1).	1+1
12	$420 = 2^2 \times 3 \times 5 \times 7$ (1); Cancel the factors other than 2 and 5: $n = 21$ (1).	1+1
13	$\frac{1}{x} = \frac{6-\sqrt{35}}{6^2-35} = 6 - \sqrt{35}$ (1); $x + \frac{1}{x} = 12$ (1).	1+1
14	$0.\overline{29} = \frac{29}{99}$, $0.\overline{92} = \frac{92}{99}$ (1); Sum $= \frac{11}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0.4\overline{7}$: $100x - 10x = 43$ (1); $x = \frac{43}{90}$ (1); Long division of p by q gives $0.4\overline{7}$ (1).	1+1+1
16	$\frac{1}{x} = 2 - \sqrt{3}$ since $2^2 - 3 = 1$, so $x + \frac{1}{x} = 4$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 16 - 2 = 14$ (1).	1+1+1
17	Suppose $3 + \sqrt{2} = r$ is rational; Then $\sqrt{2} = r - 3$ (1); $r - 3$ is rational, so $\sqrt{2}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x = 5.3\overline{2}$: $100x - 10x = 479$ (1); $x = \frac{479}{90}$ (1); Long division of p by q gives $5.3\overline{2}$ (1).	1+1+1
18	$\frac{1}{x} = 3 - \sqrt{8}$ since $3^2 - 8 = 1$, so $x + \frac{1}{x} = 6$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 36 - 2 = 34$ (1).	1+1+1
	OR: Suppose $5 + \sqrt{3} = r$ is rational; Then $\sqrt{3} = r - 5$ (1); $r - 5$ is rational, so $\sqrt{3}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Divide 3 by 7 until a remainder repeats (1); $\frac{3}{7} = 0.\overline{428571}$ (1); Block 428571 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(7 + 5) = 24$ (1); Second $= 7 - 5 = 2$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{35}$ cancel (2).	1+1+1+2
20	Divide 5 by 13 until a remainder repeats (1); $\frac{5}{13} = 0.\overline{384615}$ (1); Block 384615 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(11 + 3) = 28$ (1); Second $= 11 - 3 = 8$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{33}$ cancel (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	Cancel every prime factor other than 2 and 5.
2	**(a)** 6	$\frac{1}{x} = a - \sqrt{8}$.
3	**(a)** $\frac{15}{11}$	Write as $1 + 0.\overline{ab}$.
4	**(b)** irrational	Reason about decimal expansions.
5	**(b)** 3	Find the order of 10 modulo q.
6	**(a)** 3	Multiply the fraction by 11 and reduce.
7	**(a)** $\frac{10}{9}$	Each is $\frac{d}{9}$.
8	**(b)** 1	Add the numerators over 99.
9	**(b)** 21	Cancel every prime factor other than 2 and 5.
10	**(c)** 8	$\frac{1}{x} = a - \sqrt{15}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$90 = 2 \times 3^2 \times 5$ (1); Cancel the factors other than 2 and 5: $n = 9$ (1).	1+1
12	$\frac{1}{x} = \frac{7-\sqrt{48}}{7^2-48} = 7 - \sqrt{48}$ (1); $x + \frac{1}{x} = 14$ (1).	1+1
13	$0.\overline{36} = \frac{36}{99}$, $0.\overline{63} = \frac{63}{99}$ (1); Sum $= 1$ (1).	1+1
14	$1430 = 2 \times 5 \times 11 \times 13$ (1); Cancel the factors other than 2 and 5: $n = 143$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 6.7\overline{9}$: $100x - 10x = 612$ (1); $x = \frac{34}{5}$ (1); Long division of p by q gives 6.8 (1).	1+1+1
16	$\frac{1}{x} = 4 - \sqrt{15}$ since $4^2 - 15 = 1$, so $x + \frac{1}{x} = 8$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 64 - 2 = 62$ (1).	1+1+1
17	Suppose $7 + \sqrt{5} = r$ is rational; Then $\sqrt{5} = r - 7$ (1); $r - 7$ is rational, so $\sqrt{5}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x = 0.9\overline{4}$: $100x - 10x = 85$ (1); $x = \frac{17}{18}$ (1); Long division of p by q gives $0.9\overline{4}$ (1).	1+1+1
18	$\frac{1}{x} = 5 - \sqrt{24}$ since $5^2 - 24 = 1$, so $x + \frac{1}{x} = 10$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 100 - 2 = 98$ (1).	1+1+1
	OR: Suppose $4 + \sqrt{6} = r$ is rational; Then $\sqrt{6} = r - 4$ (1); $r - 4$ is rational, so $\sqrt{6}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Divide 4 by 41 until a remainder repeats (1); $\frac{4}{41} = 0.\overline{09756}$ (1); Block 09756 (1); Block length $= 5$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(13 + 2) = 30$ (1); Second $= 13 - 2 = 11$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{26}$ cancel (2).	1+1+1+2
20	Divide 10 by 37 until a remainder repeats (1); $\frac{10}{37} = 0.\overline{270}$ (1); Block 270 (1); Block length $= 3$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(10 + 6) = 32$ (1); Second $= 10 - 6 = 4$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{60}$ cancel (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{16}{11}$	Write as $1+0.\overline{ab}$.
2	**(c)** infinitely many rational numbers	Reason about decimal expansions.
3	**(a)** 6	Find the order of 10 modulo q.
4	**(a)** 5	Multiply the fraction by 11 and reduce.
5	**(c)** $\frac{5}{9}$	Each is $\frac{d}{9}$.
6	**(a)** $\frac{11}{9}$	Add the numerators over 99.
7	**(a)** 11	Cancel every prime factor other than 2 and 5.
8	**(d)** 10	$\frac{1}{x}=a-\sqrt{24}$.
9	**(b)** $\frac{18}{11}$	Write as $1+0.\overline{ab}$.
10	**(c)** irrational	Reason about decimal expansions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{x}=\frac{8-\sqrt{63}}{8^2-63}=8-\sqrt{63}$ (1); $x+\frac{1}{x}=16$ (1).	1+1
12	$0.\overline{58}=\frac{58}{99}$, $0.\overline{85}=\frac{85}{99}$ (1); Sum $=\frac{13}{9}$ (1).	1+1
13	$504=2^3\times3^2\times7$ (1); Cancel the factors other than 2 and 5: $n=63$ (1).	1+1
14	$\frac{1}{x}=\frac{9-\sqrt{80}}{9^2-80}=9-\sqrt{80}$ (1); $x+\frac{1}{x}=18$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x=7.1\overline{8}$: $100x-10x=647$ (1); $x=\frac{647}{90}$ (1); Long division of p by q gives $7.1\overline{8}$ (1).	1+1+1
16	$\frac{1}{x}=6-\sqrt{35}$ since $6^2-35=1$, so $x+\frac{1}{x}=12$ (1); $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$ (1); $=144-2=142$ (1).	1+1+1
17	Suppose $2+\sqrt{7}=r$ is rational; Then $\sqrt{7}=r-2$ (1); $r-2$ is rational, so $\sqrt{7}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x=8.2\overline{3}$: $100x-10x=741$ (1); $x=\frac{247}{30}$ (1); Long division of p by q gives $8.2\overline{3}$ (1).	1+1+1
18	$\frac{1}{x}=7-\sqrt{48}$ since $7^2-48=1$, so $x+\frac{1}{x}=14$ (1); $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$ (1); $=196-2=194$ (1).	1+1+1
	OR: Suppose $9+\sqrt{10}=r$ is rational; Then $\sqrt{10}=r-9$ (1); $r-9$ is rational, so $\sqrt{10}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Divide 7 by 27 until a remainder repeats (1); $\frac{7}{27}=0.\overline{259}$ (1); Block 259 (1); Block length $=3$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $=2(17+8)=50$ (1); Second $=17-8=9$ (1); Both are rational (1); In the first, the cross terms $\pm2\sqrt{136}$ cancel (2).	1+1+1+2
20	Divide 12 by 101 until a remainder repeats (1); $\frac{12}{101}=0.\overline{1188}$ (1); Block 1188 (1); Block length $=4$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $=2(19+7)=52$ (1); Second $=19-7=12$ (1); Both are rational (1); In the first, the cross terms $\pm2\sqrt{133}$ cancel (2).	1+1+1+2

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